DETECTING PRETRAINING DATA FROM LARGE LANGUAGE MODELS

Weijia Shi[1] * **Anirudh Ajith**[2*] **Mengzhou Xia**[2] **Yangsibo Huang**[2]
Daogao Liu[1] **Terra Blevins**[1] **Danqi Chen**[2] **Luke Zettlemoyer**[1]
[1]University of Washington [2]Princeton University
swj0419.github.io/detect-pretrain.github.io

arXiv:2310.16789v3 [cs.CL] 9 Mar 2024

ABSTRACT

Although large language models (LLMs) are widely deployed, the data used to train them is rarely disclosed. Given the incredible scale of this data, up to trillions of tokens, it is all but certain that it includes potentially problematic text such as copyrighted materials, personally identifiable information, and test data for widely reported reference benchmarks. However, we currently have no way to know which data of these types is included or in what proportions. In this paper, we study the pretraining data detection problem: *given a piece of text and black-box access to an LLM without knowing the pretraining data, can we determine if the model was trained on the provided text?* To facilitate this study, we introduce a dynamic benchmark WIKIMIA that uses data created before and after model training to support gold truth detection. We also introduce a new detection method MIN-K% PROB based on a simple hypothesis: an unseen example is likely to contain a few outlier words with low probabilities under the LLM, while a seen example is less likely to have words with such low probabilities. MIN-K% PROB can be applied without any knowledge about the pretraining corpus or any additional training, departing from previous detection methods that require training a reference model on data that is similar to the pretraining data. Moreover, our experiments demonstrate that MIN-K% PROB achieves a 7.4% improvement on WIKIMIA over these previous methods. We apply MIN-K% PROB to three real-world scenarios, copyrighted book detection, contaminated downstream example detection and privacy auditing of machine unlearning, and find it a consistently effective solution.

1 INTRODUCTION

As the scale of language model (LM) training corpora has grown, model developers (e.g, GPT-4 (Brown et al., 2020a) and LLaMA 2 (Touvron et al., 2023b)) have become reluctant to disclose the full composition or sources of their data. This lack of transparency poses critical challenges to scientific model evaluation and ethical deployment. Critical private information may be exposed during pretraining; previous work showed that LLMs generated excerpts from copyrighted books (Chang et al., 2023) and personal emails (Mozes et al., 2023), potentially infringing upon the legal rights of original content creators and violating their privacy. Additionally, Sainz et al. (2023); Magar & Schwartz (2022); Narayanan (2023) showed that the pretraining corpus may inadvertently include benchmark evaluation data, making it difficult to assess the effectiveness of these models.

In this paper, we study the pretraining data detection problem: given a piece of text and black-box access to an LLM with no knowledge of its pretraining data, can we determine if the model was pretrained on the text? We present a benchmark, WIKIMIA, and an approach, MIN-K% PROB, for pretraining data detection. This problem is an instance of Membership Inference Attacks (MIAs), which was initially proposed by Shokri et al. (2016). Recent work has studied *fine-tuning* data detection (Song & Shmatikov, 2019; Shejwalkar et al., 2021; Mahloujifar et al., 2021) as an MIA problem. However, adopting these methods to detect the pertaining data of contemporary large LLMs presents two unique technical challenges: First, unlike fine-tuning which usually runs for multiple epochs, pretraining uses a much larger dataset but exposes each instance only once, significantly

*Equal contribution

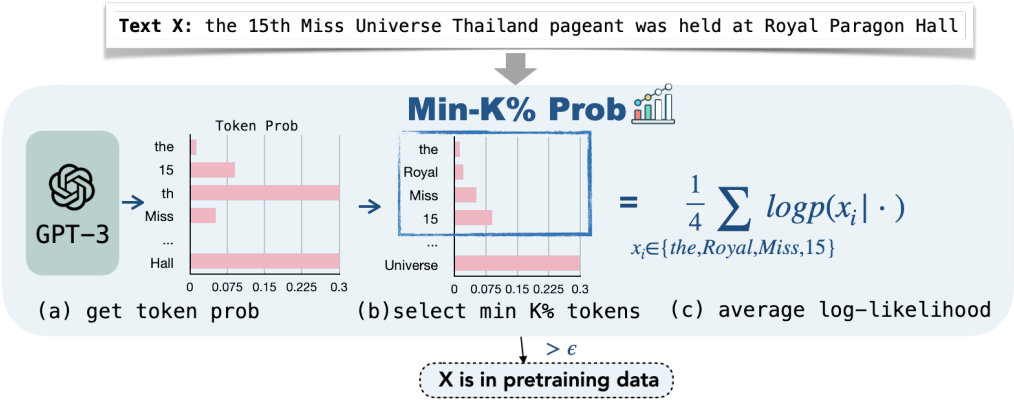

Figure 1: **Overview of MIN-K% PROB**. To determine whether a text X is in the pretraining data of a LLM such as GPT, MIN-K% PROB first gets the probability for each token in X, selects the $k\%$ tokens with minimum probabilities and calculates their average log likelihood. If the average log likelihood is high, the text is likely in the pretraining data.

reducing the potential memorization required for successful MIAs (Leino & Fredrikson, 2020; Kandpal et al., 2022). Besides, previous methods often rely on one or more reference models (Carlini et al., 2022; Watson et al., 2022) trained in the same manner as the target model (e.g., on the shadow data sampled from the same underlying pretraining data distribution) to achieve precise detection. This is not possible for large language models, as the training distribution is usually not available and training would be too expensive.

Our first step towards addressing these challenges is to establish a reliable benchmark. We introduce WIKIMIA, a dynamic benchmark designed to periodically and automatically evaluate detection methods on any newly released pretrained LLMs. By leveraging the Wikipedia data timestamp and the model release date, we select old Wikipedia event data as our member data (i.e, *seen* data during pretraining) and recent Wikipedia event data (e.g., after 2023) as our non-member data (*unseen*). Our datasets thus exhibit three desirable properties: (1) **Accurate**: events that occur after LLM pretraining are guaranteed not to be present in the pretraining data. The temporal nature of events ensures that non-member data is indeed unseen and not mentioned in the pretraining data. (2) **General**: our benchmark is not confined to any specific model and can be applied to various models pretrained using Wikipedia (e.g., OPT, LLaMA, GPT-Neo) since Wikipedia is a commonly used pretraining data source. (3) **Dynamic**: we will continually update our benchmark by gathering newer non-member data (i.e., more recent events) from Wikipedia since our data construction pipeline is fully automated.

MIA methods for finetuning (Carlini et al., 2022; Watson et al., 2022) usually calibrate the target model probabilities of an example using a shadow reference model that is trained on a similar data distribution. However, these approaches are impractical for pretraining data detection due to the black-box nature of pretraining data and its high computational cost. Therefore, we propose a reference-free MIA method MIN-K% PROB. Our method is based on a simple hypothesis: an unseen example tends to contain a few outlier words with low probabilities, whereas a seen example is less likely to contain words with such low probabilities. MIN-K% PROB computes the average probabilities of outlier tokens. MIN-K% PROB can be applied without any knowledge about the pretrainig corpus or any additional training, departing from existing MIA methods, which rely on shadow reference models (Mattern et al., 2023; Carlini et al., 2021). Our experiments demonstrate that MIN-K% PROB outperforms the existing strongest baseline by 7.4% in AUC score on WIKIMIA. Further analysis suggests that the detection performance correlates positively with the *model size* and *detecting text length*.

To verify the applicability of our proposed method in real-world settings, we perform three case studies: copyrighted book detection (§5), privacy auditing of LLMs (§7) and dataset contamination detection (§6). We find that MIN-K% PROB significantly outperforms baseline methods in both scenarios. From our experiments on copyrighted book detection, we see strong evidence that GPT-3 [1] is pretrained on copyrighted books from the Books3 dataset (Gao et al., 2020; Min et al., 2023). From our experiments on privacy auditing of machine unlearning, we use MIN-K% PROB

[1] `text-davinci-003`.

to audit an unlearned LLM that is trained to forget copyrighted books using machine unlearning techniques (Eldan & Russinovich, 2023) and find such model could still output related copyrighted content. Furthermore, our controlled study on dataset contamination detection sheds light on the impact of pretraining design choices on detection difficulty; we find detection becomes harder when training data sizes increase, and occurrence frequency of the detecting example and learning rates decreases.

2 PRETRAININING DATA DETECTION PROBLEM

We study pretraining data detection, the problem of detecting whether a piece of text is part of the training data. First, we formally define the problem and describe its unique challenges that are not present in prior finetuning data detection studies (§2.1). We then curate WIKIMIA, the first benchmark for evaluating methods of pretraining data detection (§2.2).

2.1 PROBLEM DEFINITION AND CHALLENGES

We follow the standard definition of the membership inference attack (MIA) by Shokri et al. (2016); Mattern et al. (2023). Given a language model f_θ and its associated pretraining data $\mathcal{D} = \{z_i\}_{i \in [n]}$ sampled from an underlying distribution $\mathbb{D}$, the task objective is to learn a detector h that can infer the membership of an arbitrary data point x: $h(x, f_\theta) \rightarrow \{0, 1\}$. We follow the standard setup of MIA, assuming that the detector has access to the LM only as a black box, and can compute token probabilities for any data point x.

Challenge 1: Unavailability of the pretraining data distribution. Existing state-of-art MIA methods for data detection during finetuning (Long et al., 2018; Watson et al., 2022; Mireshghallah et al., 2022a) typically use reference models g_γ to compute the background difficulty of the data point and to calibrate the output probability of the target language model : $h(x, f_\theta, g_\gamma) \rightarrow \{0, 1\}$. Such reference models usually share the same model architecture as f_θ and are trained on shadow data $D_{\text{shadow}} \subset \mathbb{D}$ (Carlini et al., 2022; Watson et al., 2022), which are sampled from the same underlying distribution $\mathbb{D}$. These approaches assume that the detector can access (1) the distribution of the target model's training data, and (2) a sufficient number of samples from $\mathbb{D}$ to train a calibration model.

However, this assumption of accessing the distribution of pretraining training data is not realistic because such information is not always available (e.g., not released by model developers (Touvron et al., 2023b; OpenAI, 2023)). Even if access were possible, pretraining a reference model on it would be extremely computationally expensive given the incredible scale of pretraining data. In summary, the pretraining data detection problem aligns with the MIA definition but includes an assumption that the detector has no access to pretraining data distribution $\mathbb{D}$.

Challenge 2: Detection difficulty. Pretraining and finetuning differ significantly in the amount of data and compute used, as well as in optimization setups like training epochs and learning rate schedules. These factors significantly impact detection difficulty. One might intuitively deduce that detection becomes harder when dataset sizes increase, and the training epochs and learning rates decrease. We briefly describe some theoretical evidence that inform these intuitions in the following and show empirical results that support these hypotheses in §6.

To illustrate, given an example $z \in D$, we denote the model output as $f_\theta(z)$ Now, take another example y sampled from $\mathbb{D} \setminus D$ (not part of the pretraining data). Determining whether an example x was part of the training set becomes challenging if the outputs $f_\theta(z)$ and $f_\theta(y)$ are similar. The degree of similarity between $f_\theta(z)$ and $f_\theta(y)$ can be quantified using the total variation distance. According to previous research (Hardt et al., 2016; Bassily et al., 2020), the bound on this total variation distance between $f_\theta(z)$ and $f_\theta(y)$ is directly proportional to the *occurrence frequency of the example x*, *learning rates*, and the *inverse of dataset size*, which implies the detection difficulty correlates with these factors as well.